

SARDAR PATEL UNIVERSITY
T.Y.B.C.A. EXAMINATION, VI SEM

Date : 27th Mar 2017, Monday
Session : Evening
Course No : US06FBCA01

Time : 02:00pm To 05:00pm
Sub: System Software
Total marks : 70

Q – 1 Multiple Choice Question**[10]**

- i) _____ part build the desired target program by using intermediate representation of two part.
(a) analysis (b) Synthesis (c) Execution (d) None of these
- ii) _____ attempts to discover potential bugs without running the program.
(a) Pretty printer (b) Structure Editors (c) Static checker (d) None
- iii) _____ form is also known as "Three – address code".
(a) Intermediate code generation (b) Code Generation
(c) Analysis phase (d) Code Optimization
- iv) _____ data structure is used in synthesis phase.
(a) DFD (b) Symbol Table (c) Name table (d) All of the above
- v) Mnemonic has _____ primary fields.
(a) Mnemonic and names (b) Name and address
(c) Mnemonic and opcode (d) Name and opcode
- vi) In a simple assembly language, the First operand is always _____.
(a) Register (b) Data declaration (c) Memory word (d) Mnemonic
- vii) Address of the origin assumed by the translator is known as _____.
(a) Linked Origin (b) Translated Origin (c) Load Origin (d) Origin
- viii) The _____ maintains an internal form which is used to perform the edit operation.
(a) Linker (b) Loader (c) Editor (d) None of these
- ix) _____ helps the user in selecting test data for his program..
(a) Test Data Generator (b) Language Processor
(c) Software Tools (d) Program Environment
- x) _____ helps in obtaining information for localization of errors..
(a) Profile monitors (b) Debug Monitors (c) Chosen text (d) All of above

Q – 2 Short Answer attempt any ten (Each carry 2 marks)**[20]**

- i) Draw a diagram of a language processing system.
- ii) Define Application Domain and Execution Domain
- iii) What are structure editor and pretty printer?
- iv) List four advantage of Assembly Language
- v) What is machine instructions format?
- vi) Explain <Operand Spec> with its syntax.
- vii) Explain the address sensitive program
- viii) Draw the schematic diagram of program execution.
- ix) Difference between Linking and Loading
- x) List down fundamental steps in program development
- xi) Explain Hypertext, Hyper Card and Menulay
- xii) List out the facilities provide by debug monitors for dynamic debugging.

Q – 3 Explain language processing activity in detail [10]

OR

Q – 3 Explain different phases of compiler in detail. [10]

Q – 4 a) Explain design specification of an assembler [06]

b) Write table of mnemonic opcode for machine instructions. [04]

OR

Q – 4 a) Explain elements of Assembly language programming in detail [06]

b) List and explain types of assembly language statements. [04]

Q – 5 a) Explain Editors in details. [06]

b) Explain Relocation and linking concept [04]

OR

Q – 5 a) Explain Translated, Linked and Load Time address. [06]

b) Explain design of Editors [04]

Q – 6 a) Discuss software tools for programming development. [06]

b) Explain Debug Monitor [04]

OR

Q – 6 a) Explain structure of user interface and User interface management systems [06]

b) Explain Program environments in brief [04]

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